

2016.0 RANGE ROVER (LG), 501-00

BODY SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.

2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL

- Incorrectly aligned door(s)
- Obstruction within door aperture or door latch
- Door latches and strikers
- Door frame seal(s)
- Door aperture seal(s)
- Door hinges

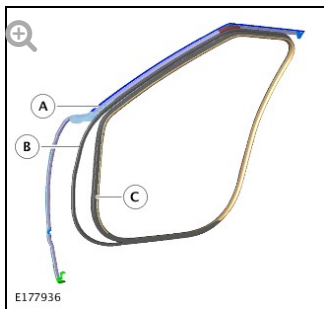
3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

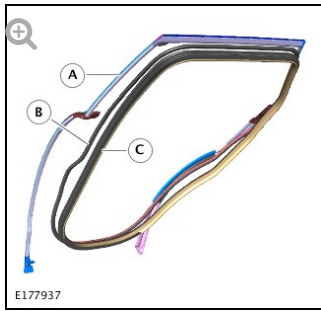
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Squeak/creak from the front/rear door(s)	- Obstruction within door aperture or door latch - Incorrect door striker-Alignment/adjustment - Door frame seal coating - Worn/damaged door frame seal(s) - Worn/damaged door aperture seal(s) - Paint wear	- Remove the obstruction within the door aperture or latch as necessary - Check/adjust the door striker as necessary - GO to Pinpoint Test A. - Check and replace the worn or damaged door frame seals as necessary - Check and replace the worn or damaged door aperture seals as necessary - Check and repair the door frame/aperture paint damage as necessary

PINPOINT TESTS

PINPOINT TEST A : FRONT DOOR	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: FRONT DOOR FRAME - MARGIN SEAL	
	1. Obtain a complete description of the customer concerns or suspected condition - Front Door - Seal A - Margin Seal (outer door frame seal) - Seal B - Door Frame Weatherstrip - Primary Seal (this seal is bonded to the door) - Seal C - Secondary Seal (this seal is mounted to the door aperture)
	2. Verify the customer concern by carrying out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs - Rear Door - Seal A - Margin Seal (outer door frame seal)



- Seal B - Door Frame Weatherstrip - Primary Seal (this seal is bonded to the door)
- C - Secondary Seal (this seal is mounted to the door aperture)



- 3 Try to establish if the noise is coming from the front or rear door(s), above or below the waist line?
- Arrow Denotes Waistline

Is the noise coming from the front door(s)
Yes
 Front door noise **GO to A2.**
No
 Rear door noise **GO to A7.**

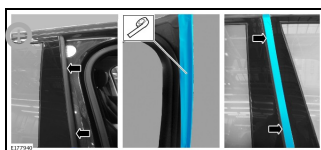
A2: FRONT DOOR FRAME ALIGNMENT - MARGIN SEAL



- 1 Check the front and rear above waistline door frame B pillar finisher alignment, the front B pillar finisher to rear B finisher should be flush $\pm 1.5\text{mm}$. The opening gap between both finishers is $5\text{mm} \pm 1.5\text{mm}$

If the noise coming from the front door frame margin seal.
 Is the door striker/alignment correct?
Yes
GO to A3.
No
 If the B pillar alignment is not correct, adjust the door striker. Road test the vehicle to confirm if the noise has stopped

A3: FRONT DOOR FRAME - MARGIN SEAL

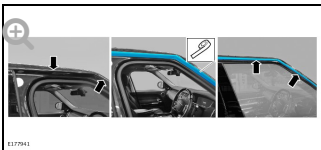


- 1 Apply suitable tape to the inner face of the front door frame finisher, on the area which contacts the rear door frame margin seal

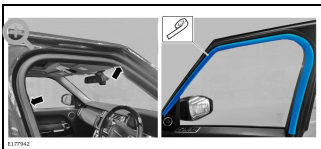
- 2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs

	Has the tape stopped the noise coming from the rear door frame margin seal? Yes If the tape has stopped the noise, apply four coats of AC602/2 to the rear door frame margin seal, allowing drying time between coats Remove the tape from the front door finisher and clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite rear door frame margin seal No GO to A4.
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A4: FRONT DOOR FRAME - MARGIN TOP SEAL

	1 Apply suitable tape to the front body aperture, on the area which contacts the front door frame top margin seal
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the front door frame top margin seal? Yes Apply coating AC612/21 (liquid) 50ml pump spray to the flock coating on the front door frame top margin seal Remove the tape from the front body aperture and clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite front door frame top margin seal No GO to A5.

A5: FRONT DOOR APERTURE - SECONDARY SEAL

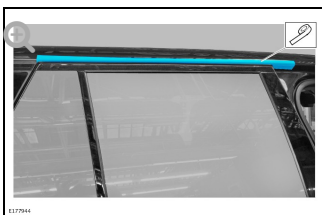
	1 Apply suitable tape to the front door frame inner trim, on the area which contacts the front door aperture secondary seal
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the front door aperture secondary seal? Yes Apply four coats of AC602/2 to the front door aperture secondary seal allowing drying time between coats Remove the tape from the front door frame inner trim and

	clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite front door aperture secondary seal No GO to A6.
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A6: FRONT DOOR WEATHERSTRIP - PRIMARY SEAL (THIS SEAL IS BONDED TO THE DOOR)

	1 Apply suitable tape to the front door aperture, on the area which contacts the front door frame weatherstrip
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the front door weatherstrip? Yes Apply four coats of AC602/2 to the front door frame weatherstrip allowing drying time between coats Remove the tape from the front door frame aperture and clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite front door frame weatherstrip No The front door squeak/creak checks are now complete, If the noise is still evident, suspect the rear door(s) GO to A7.

A7: REAR DOOR FRAME - MARGIN TOP SEAL

	1 Apply suitable tape to the rear door top aperture, on the area which contacts the rear door frame top margin seal
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the rear door frame top margin seal? Yes Apply coating AC612/21 (liquid) 50ml pump spray to the flock coating on the rear door frame top margin seal Remove the tape from the rear door aperture and clean.

	Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite rear door frame top margin seal No GO to A8.
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A8: REAR DOOR APERTURE - SECONDARY SEAL

	1 Apply suitable tape to the rear door frame inner trim, on the area which contacts the rear door aperture secondary seal
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the rear door aperture secondary seal? Yes Apply four coats of AC602/2 to the rear door aperture secondary seal allowing drying time between coats Remove the tape from the rear door frame inner trim and clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the opposite rear door aperture secondary seal No GO to A9.

A9: REAR DOOR WEATHERSTRIP - PRIMARY SEAL (THIS SEAL IS BONDED TO THE DOOR)

	1 Apply suitable tape to the rear door aperture, on the area which contacts the rear door frame weatherstrip
	2 Carry out a road test, as closely as possible reproducing the drive/road conditions under which the noise occurs
	Has the tape stopped the noise coming from the rear door weatherstrip? Yes Apply four coats of AC602/2 to the rear door frame weatherstrip allowing drying time between coats Remove tape from the rear door frame aperture and clean. Road test the vehicle to confirm if the noise has stopped At this point it is advisable to apply the coating, to the

opposite rear door frame weatherstrip

No

The side door squeak/creak tests are now complete. If the noise is still evident, it may be necessary to repeat some tests under different road conditions